BLIND ASSISTANT ROBOT — SYSTEM BLOCK DIAGRAM

Integrated Subsystems Architecture: Mechanical, Actuation/Power, Electronic Controllers (Arduino UNO Q Dual-Brain), & Firmware/Logic Flow

🔲 3. ELECTRONIC CONTROLLERS — ARDUINO UNO Q DUAL-BRAIN SBC

🐧 Linux MPU Brain: Qualcomm® Dragonwing™ QRB2210

Quad-Core ARM Cortex-A53 @ 2.0 GHz | Adreno 702 GPU | 2GB/4GB LPDDR4 | Debian Linux

SLAM Toolbox
Synchronous 2D Occupancy Grid (/map)
Resolution: 0.05m / cell | Scan Matcher

ROS 2 Middleware Stack
Humble / Jazzy | TF2 Transform Buffer
map -> odom -> base_footprint -> sensors

Semantic Vision Node
OpenCV HSV Landmark Contours (7 Classes)
Central 50% ROI Depth Extraction
Adreno 702 GPU Accelerated

HMI & Terminal Interfaces
Native OpenCV SLAM Map (700x700 + HUD)
Live Camera View (640x480)
room_cli (NLP Synonym Intent Parser)

Autonomous Explorer Engine
Frontier Detection & BFS Clustering
8-Connected A* Planner + String Pulling
Doorway Corridor Funneling (+/-14° cone)
Dynamic Pedestrian Yielding (< 1.25m)

Regulated Pure Pursuit
Adaptive Lookahead (0.25 - 0.48m)
Curvature Speed Scaling & Slew Limiter
Generates /cmd_vel (v, omega)

⚡ ON-CHIP ARDUINO BRIDGE RPC LAYER
High-Speed Shared Memory / Virtual Serial (/dev/ttyRPMSG) | Zero Cable Latency

⚡ Real-Time MCU Brain: STMicroelectronics® STM32U585

32-Bit ARM Cortex-M33 @ 160 MHz | Zephyr RTOS / Arduino Runtime | Sub-Microsecond Determinism

Hardware Interrupt Handlers (ISRs)
EXTI2 (Pin 2: Left) & EXTI3 (Pin 3: Right)
20 Hz Formatted Tick Packet Broadcast

Safety Watchdog Failsafe
500ms Timeout: Auto-Zero Motor PWM
Runaway Motion Prevention

32-Bit Hardware PWM Timers
Pins 5 & 6 @ 980 Hz (0 - 255 Duty)
Direction Gates (Pins 7, 8, 9, 10)

📡 SENSOR SUITE & ASSISTIVE PERIPHERALS

Audio DAC & Speaker
Assistive Natural Language Voice Speech
Spoken Alerts via /voice_speech

YDLidar X4 Planar Scanner
360° Planar Laser Turret @ 10Hz
360 Samples (1.0° Res) | Range: 0.12m - 10.0m

Intel RealSense D435i
RGB-D Depth Perception Camera
640x480 RGB @ 15Hz | 32FC1 Depth
HFOV: 1.15 rad (65.9°) | Range: 0.10m - 10.0m

⚡ 2. ACTUATION & POWER SUBSYSTEM

3S Li-ion Battery Pack
11.1V - 12.6V | 4500 mAh (25C)
Runtime: 3.0 - 4.0 Hours

Protection & Switch
10A Blade Fuse + Rocker Kill Switch

DC-DC Buck Converter
LM2596 / XL4015 (12V -> 5.0V @ 3A)

Dual H-Bridge Driver
L298N / TB6612FNG (2A / Channel)
Hardware PWM Speed & Dir Control

Left 12V DC Gearmotor
1:30 Metal Spur Gearbox | 40 Nm Stall

Right 12V DC Gearmotor
1:30 Metal Spur Gearbox | 40 Nm Stall

Left Optical Encoder
360 Counts / Revolution (CPR)

Right Optical Encoder
360 Counts / Revolution (CPR)

🛠 1. MECHANICAL STRUCTURE & KINEMATICS

Differential Drive Chassis Base
SolidWorks CAD Aluminum/Acrylic Baseplate (260x190x76 mm)
Total Mass: ~2.3 kg (CoM z = +0.032m) | Footprint Radius: 0.22m

Dual Drive Wheels
Diameter: 66mm | Rubber Tires (mu = 1.2)
Track Width (Separation): L = 176mm

Ground Contact Supports
Rear 1.5-in Nylon Omni Caster (x = -0.10m, mu = 0.001)
Front Anti-Tip Skid Ball (x = +0.10m, mu = 0.001)

Rigid Sensor Mounts
Top Turret: Elevated LiDAR (z = +0.081m, 360° Clear Line-of-Sight)
Front Aperture: RealSense Recessed Bracket (x = +0.062m, z = +0.052m)

/map (OccupancyGrid) & TF (map->odom)
USB-UART @ 128,000 baud
(/scan 360° LaserScan)
USB 3.0 SuperSpeed
(/camera/image_raw & /camera/depth/image_raw)
5.0V Turret Motor Power
Synchronous Costmap & Robot Pose
3D Discovered Landmarks & /detected_human
/guide_to & CLI intents
/voice_speech (Spoken Audio Alerts)
Corridor Ray Checks
Wheel Odometry (/odom) & odom->base_link TF
Active A* Trajectory
5V USB-C VBUS
Hardware Ext Interrupt (Pin 2: EXTI2)
Hardware Ext Interrupt (Pin 3: EXTI3)
/cmd_vel (Linear v, Angular omega)
High-Speed Shared Memory (20Hz Encoder Ticks)
Watchdog Reset if > 500ms
Internal RPC Velocity Targets
12.6V Main
12V Input
12V High-Current Bus
12V PWM Drive
12V PWM Drive
Motor Shaft
Motor Shaft
Direct Drive Left
Direct Drive Right
5.0V @ 3A Main Logic Rail (Powers UNO Q)
Hardware PWM (Pins 5,6) & Dir (Pins 7,8,9,10)